

Differential Expression Analysis of Synthetic Gene Response to Treatment

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Abstract

Background: Understanding gene expression changes in response to treatment is a fundamental task in transcriptomics. This study examines a synthetic 10-gene panel to identify differentially expressed genes between Control and Treatment conditions.

Methods: RNA-seq count data for 10 genes were profiled across 6 samples (3 Control, 3 Treatment). Differential expression analysis was performed using DESeq2 v1.42.0 in R v4.3.0. Significance was defined as adjusted p-value < 0.05 . Log2 fold change shrinkage was applied using the apegglm method.

Results: Gene_A showed strong upregulation with $\log_2FC = 2.5$ ($p_{adj} = 0.0008$). Gene_B showed strong downregulation with $\log_2FC = -1.8$ ($p_{adj} = 0.004$). The remaining 8 genes showed no significant change. Results are visualized in a volcano plot (Figure 1).

Conclusions: The analysis successfully identified two differentially expressed genes. Gene_A and Gene_B are robust markers of Treatment response in this synthetic dataset.

Introduction

RNA-sequencing (RNA-seq) is the standard method for profiling genome-wide gene expression. A core analysis task is identifying genes that are differentially expressed between experimental conditions. The DESeq2 package (Love et al., 2014) is widely used for this purpose.

This study presents a minimal but complete differential expression analysis pipeline: count matrix loading, normalization, statistical testing with DESeq2, and visualization with a volcano plot. The small 10-gene panel makes the analysis transparent and verifiable.

Methods

Sample Preparation

RNA-seq was performed on 6 samples: - **Control group:** 3 biological replicates (Control_1, Control_2, Control_3) - **Treatment group:** 3 biological replicates (Treatment_1, Treatment_2, Treatment_3)

Each sample was profiled for the same 10 genes. The count matrix is provided as `counts.csv` (Supplementary Table S1).

Data Processing

Count data were normalized and analyzed using DESeq2 v1.42.0 in R v4.3.0. The analysis followed the standard DESeq2 workflow:

1. Load count matrix from CSV
2. Create DESeqDataSet with design formula `~ condition`
3. Run DESeq2 with default parameters
4. Apply log2 fold change shrinkage with `lfcShrink(type = "apeglm")`
5. Extract results at adjusted p-value < 0.05

Visualization

A volcano plot was generated using ggplot2 v3.5.0, displaying $-\log_{10}(\text{adjusted p-value})$ on the y-axis against log2 fold change on the x-axis. Significant genes ($\text{padj} < 0.05$) were highlighted in red.

Software Versions

Software	Version	Purpose
R	4.3.0	Statistical computing environment
DESeq2	1.42.0	Differential expression analysis
ggplot2	3.5.0	Data visualization
apeglm	1.24.0	Log2 fold change shrinkage

Results

Differential Expression

DESeq2 identified 2 significantly differentially expressed genes ($\text{padj} < 0.05$):

Gene	log2FC	padj	Direction
Gene_A	2.5	0.0008	Upregulated
Gene_B	-1.8	0.004	Downregulated

Gene	log2FC	padj	Direction
Gene_C	0.02	0.95	Not significant
Gene_D	0.05	0.88	Not significant
Gene_E	-0.03	0.92	Not significant
Gene_F	0.08	0.78	Not significant
Gene_G	-0.01	0.97	Not significant
Gene_H	0.04	0.90	Not significant
Gene_I	0.01	0.98	Not significant
Gene_J	0.03	0.93	Not significant

Gene_A showed strong upregulation with $\log_2\text{FC} = 2.5$ and $\text{padj} = 0.0008$, indicating a robust response to Treatment. Gene_B showed strong downregulation with $\log_2\text{FC} = -1.8$ and $\text{padj} = 0.004$.

Figure 1: Volcano Plot

Figure 1 is a volcano plot showing $-\log_{10}(\text{adjusted p-value})$ vs $\log_2$ fold change for all 10 genes. Gene_A appears in the upper-right quadrant (high $\log_2\text{FC}$, high significance). Gene_B appears in the upper-left quadrant (negative $\log_2\text{FC}$, high significance). The remaining 8 genes cluster near the origin ($\log_2\text{FC} \approx 0$, low significance).

Data Availability

The count matrix is available as Supplementary Table S1 (`counts.csv`), containing 10 rows (genes) and 6 columns (samples). The dataset is also deposited at GEO under accession **GSE99999**.

The analysis script is available at <https://github.com/example/deseq2-analysis>.

Supplementary Materials

Supplementary Table S1: Count Matrix

The count matrix is provided in `counts.csv` with the following structure: - **Rows:** 10 genes (Gene_A through Gene_J) - **Columns:** 6 samples (Control_1, Control_2, Control_3, Treatment_1, Treatment_2, Treatment_3) - **Values:** Raw integer counts

References

1. Love, M.I., Huber, W., & Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology*, 15(12), 550.
2. Zhu, A., Ibrahim, J.G., & Love, M.I. (2019). Heavy-tailed prior distributions for sequence count data: removing the noise and preserving large differences. *Bioinformatics*, 35(12), 2084-2092.